RESEARCH-ARTICLE

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Abstract

Sport climbing, a discipline demanding high levels of muscular strength, endurance, and cognitive planning, has gained considerable popularity in recent years. The importance of managing muscle fatigue during climbing, which can substantially impair performance and potentially lead to injury, has not yet been thoroughly investigated. Predicting muscle fatigue can help to create tailored training programs and refine climbing strategies to optimise performance and safety. This study aimed to monitor and predict muscle fatigue during climbing. We conducted a multi-modal experiment involving 20 climbers, measuring their muscle activity and fatigue via electromyography (EMG) and tracking their climbing trajectories through video recordings. We compared different linear autoregressive process (AR) methods and machine learning methods that predict muscle fatigue up to 5 seconds into the future. We also successfully proposed to extend the AR model to account for expected muscle fatigue as a function of distance climbed. While this extension improved prediction over the standard AR model, with a Root mean square error (RMSE) of 16.05 (±28.42%), the non-linear Multilayer perceptron (MLP) and Gradient boosting (GB) models outperformed linear methods, with a lower RMSE of 15.09 (±27.94%)and 15.09 (±29.02%), respectively. Despite their lower accuracy compared to non-linear models, the simplicity of calculating linear models could enable real-time predictions on wearable devices, providing climbers with valuable, immediate feedback on muscle fatigue. This capability can significantly impact climbing research by providing practical tools for real-world applications, improving climbers' decision-making and enhancing safety and performance.

CCS Concepts

• **Applied computing** → **Forecasting**; *Systems biology*; • **Computing methodologies** → *Machine learning*.

Keywords

Muscle fatigue, Climbing, Sensors, Time series

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1 Introduction

Sport climbing has gained popularity recently and made its Olympic debut in 2020. Climbing performance requires power, strength, endurance, balance, and flexibility [19]. During top rope climbing, climbers climb secured by a rope attached in an anchor at the top of a route. In lead climbing, climbers protect themselves as they ascend the route by clipping their rope into points attached to the wall that are spaced apart at separate intervals. Previous research suggests that lead climbing is more mentally and physically demanding than top rope climbing [9].

Moreover, climbing involves the cognitive task of route planning, an integral part of a successful ascent. Climbers have reported that route planning allows them to link different movement sequences, avoid ineffective movements and adapt their movements to the demands of the route [22]. The chosen sequence should increase the movement flow of the climber and limit fatigue. In particular, climbers must constantly manage fatigue during the ascent by adjusting their speed, shaking out their arms to help restore blood flow, or adopting resting positions. Effectively tackling fatigue is crucial, as it significantly impacts performance and can lead to

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muscle overuse injuries, highlighting the importance of strategic fatigue management in climbing. So far, little attention has been paid to quantitatively analysing muscle fatigue over the course of a
on the wall. Closing this research gap and understanding these dynamics can lead to more effective training plans, better climbing strategies, and a lower risk of injury.

In an effort to address this gap, we present an experiment in which we recorded muscle activity in terms of EMG of the flexor digitorum superficialis (FDS) on the right forearm during climbing. In total, 20 moderate level climbers climbed the same route twice, once as a lead climb and once on top rope. While in top rope climbing the rope is secured from the top anchor, in lead climbing the climber has to clip the rope into the quickdraw attached to the wall during the ascent, which forces him to temporarily take one hand off the wall. EMG measures muscle activity and fatigue over the course of the climb. Additionally, we video recorded each climbing attempt. Based on the video recording we derived the trajectory of the climber on the wall. The trajectory was used to map muscle fatigue to the climbers' position on the wall.

Based on the climbers' limb movement, muscle activation, and position on the wall, we proposed mathematical models to predict muscle fatigue. Such predictions will help us to understand where on the route a climber will accumulate the most fatigue and where the FDS will be most activated in the future. The predictive model allows for a detailed analysis of muscle fatigue and thus the identification of critical areas on climbing routes that may require strategic planning or adaptation. By modelling fatigue in relation to specific climbing positions, climbers can also develop more effective training plans and climbing strategies. This approach can also help to reduce the risk of injury by identifying muscle overuse.

In short, our research offers several key contributions:

- We conducted a multi-modal experiment, which combines sensor data and a video trajectory of 20 participants climbing on lead and on top rope.
- We derived the climbers' trajectories on the route from the videos.
- We compared linear AR models with non-linear neural networks and ensemble methods.
- For the prediction of muscle fatigue, we improved the linear AR by including the expected muscle fatigue given trajectory height as a feature.

2 Muscle fatigue during climbing

Muscle fatigue is a local and complex phenomenon that involves a temporary reduction in the ability to carry out physical activities [7, 10]. Fatigue is caused by a series of changes in the muscles related to metabolism, structure, and energy, mainly because the muscles are not getting enough oxygen and nutrients [7]. Muscle fatigue can be the consequence of different physiological mechanisms, which makes it difficult to identify specific causes [10]. Muscle fatigue itself however can be measured by continuously monitoring myoelectric activity of a specific muscle. A non-invasive method to measure myoelectric activity is surface electromyography (sEMG). sEMG involves placing an electrode on the skin to passively measure the electrical activity of the muscles beneath. The electrical activity of a muscle reflects contraction of many muscle fibres. Muscle contraction increases the arterial blood pressure which decreases the blood flow to the working muscle. This decreases the supply of oxygen to the muscle and induces fatigue [11, 27]. With a decreased blood supply, the metabolic removal is affected which leads to a built up of lactic acid in the muscle. This can give the "burn" sensation or the experience of "pump" in which the muscle swells. The change in acidity slows the speed with which electrical signals travel through the muscle [7].

Dehyle et al. analysed the importance of different muscle groups for indoor rock climbing. They concluded that the finger flexors, specifically the FDS and flexor digitorum profundus, as well as the elbow flexors, including the biceps brachii, brachioradialis, and brachialis, play a significant role in climbing [8]. Numerous studies have highlighted the importance of the FDS and or elbow flexors in climbing. Specifically, Koukoubis et al. demonstrated that during hanging and pull-ups, the muscle activity measured by the integrated EMG (iEMG) of the FDS and brachioradialis muscles is notably high [17]. The iEMG measures the cumulative electrical activity of muscles over time, providing a quantitative assessment of muscle activation. Muscle fatigue measured by EMG occurs twice as rapidly in non-climbers compared to climbers, suggesting that training can slow fatigue [26].

Watts et al. discovered that the amplitude of the iEMG during climbing significantly exceeds that during hand grip dynamometer measurements [29]. These findings emphasize the importance of measuring muscle activity with EMG during climbing as measuring before and after climbing can lead to bias. Watts et al. examined the impact of repeated hangboard trials and different recovery times on peak and iEMG. They found that with one-minute recovery, the iEMG decreases with more hangs. However, with three minutes of recovery, the iEMG plateaus after the third repetition [28].

Vigouroux et al. conducted a study investigating muscle activity during pull-ups with various grip types. Using EMG to measure muscle activity, they found that thinner grips resulted in intensified muscle fatigue, suggesting that grip types significantly influence fatigue development during climbing activities [25]. Gou et al. recorded sEMG during and assessed blood lactate levels before and after several speed climbing runs. They examined the percentage change in the median frequency of sEMG from the start to the end of each climb [12]. Results revealed a consistent decrease in the median frequency of the sEMG after each run. They also identified a correlation between climbing time and blood lactate level. Furthermore, their research confirms Dehyle et al.'s findings, which indicated significant changes in the BB and FDS muscles [8, 12]. The literature has thoroughly examined the notion that the FDS and BB muscles experience fatigue during climbing and that correlations exist between blood lactate levels and various grip depths.

While research has emphasised the importance of FDS in climbing and the effects of grip conditions and physical loading patterns on muscle fatigue, a comprehensive, temporal analysis of muscle fatigue in climbing is lacking in the existing literature. Although insightful, these studies have largely focused on isolated aspects, leaving a critical gap in our understanding of how muscle fatigue develops and fluctuates over the course of a climbing ascent. This highlights the need for a more holistic approach that tracks and analyses muscle fatigue in real time, as this could provide valuable

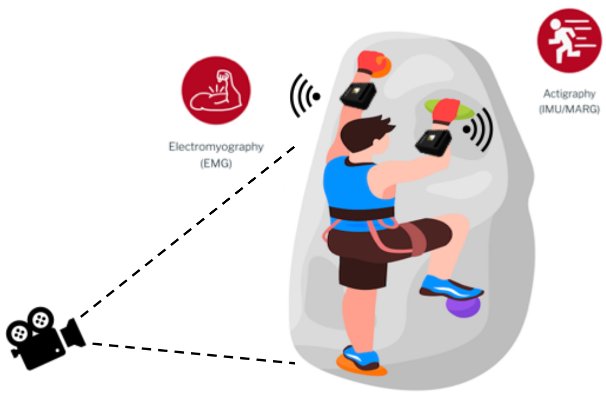


Figure 1: Graphical illustration of the multi-modal experiment, which combines video and sensor data. The sensors include the recording of several physiological signals such as Internal measurement unit (IMU) and EMG.

insights to optimise climbing strategies, improve training regimes and reduce injury risk. Our study aims to fill this gap by proposing mathematical models that predict muscle fatigue based on limb movement, muscle activation and position on the wall.

3 Climbing experiment

Figure 1 illustrates graphically the multi-modal experiment design. The climbers were instructed to climb the same route twice, once on lead and once on top rope. The climbers wore sensors on the FDS on their right forearm to measure their limb movement and muscle activity during the climb. Additionally, we recorded video during each climb. The experiment unfolded in multiple stages. This subsection will outline the procedure in the order it was conducted.

3.1 Briefing and warm-up

Participants were first informed about the experiment's objectives, data recording procedures, and their rights concerning personal information. All participants conducted a warm-up routine. The warm-up routine lasted approximately 5 minutes and included a series of exercises specifically chosen to target various muscle groups and enhance joint mobility. It began with tendon gliding exercises to promote flexibility in the hands and wrists, followed by fist squeezes to warm up the forearms. Participants then performed 20 jumping jacks to increase their heart rate and improve overall cardiovascular readiness. This was followed by arm circles to loosen the shoulder joints and prepare the upper body for movement. Participants then executed 10 repetitions of bird-dogs, focusing on core stabilisation and balance. Finally, the warm-up concluded with leg swings to dynamically stretch the leg muscles and hip joints, ensuring that participants were fully prepared to engage in the experimental activities safely and effectively.

3.2 Randomisation

Participants climbed the same route using both lead and top rope climbing styles. The order was randomised for each participant to mitigate potential confounding factors, such as accumulated fatigue and prior knowledge of the route.

3.3 Climbing

The participants and the experimenter conducted safety checks before each climb. The climbing ascent was video recorded. The participants climbed both lead and top rope. To make the top rope condition more matched to the lead condition, participants were instructed to unclip their rope from each quickdraw up to the anchor during the ascent. This method mimics the one-handed movement of lead climbing, although it does not replicate the additional challenge of pulling the rope. This is also a normal method of top roping when top roping outdoors.

3.4 Sensor

The BioPoint™ sensor is a compact biosensor device the size of a smartwatch that is equipped with several high-quality sensors, including electrocardiogram (ECG), electrodermal activity (EDA), photoplethysmography (PPG), skin temperature, a six-axis IMU and EMG. The device has 4 GB of internal memory and was suitable for the data collection [24]. The sensors were placed on both participants' right forearm on the FDS muscle, at $\frac{3}{4}$ distance from the elbow to the wrist, slightly towards the ulnar side. The instructor confirmed the placement of the sensor by palpating the muscle while the participants flexed their fingers and applied pressure with their palm to help identify the muscle. The IMU captures the right arm's motion as acceleration in the three spatial dimensions (x, y, z). The EMG sensor records surface myoelectric activity which is further introduced in Section 2.

4 Trajectory tracking

The climbers' trajectories were extracted using video analysis center of gravity (COG). The centre of gravity is of crucial importance when climbing as it determines the weight distribution and therefore affects the muscle activation of the gripping arms. The process involved capturing video footage of the climber and to manually annotate the COG coordinates frame-by-frame. A graphical user interface was developed and employed to manually annotate the COG by clicking on the climber's position. The annotated coordinates were recorded in real-time and visualised on the frame to confirm accuracy. These coordinates were then compiled into a CSV file, documenting the COG for each frame. Due to the small number of 40 videos, we considered the manual annotation as feasible, more robust and precise than automated tracking methods. Figure 2a shows the annotated trajectory on the original frame. The video were taken from the bottom with an upwards facing angle. Accordingly, we applied a perspective transformation, also known as a homography, to project the trajectory onto a 2D plane. The dimensions of the wall section are known with 1.5 meter width and 19.3m of height. The projective transformation maps a quadrilateral in one plane to another quadrilateral in a different plane. It preserves straight lines but not necessarily parallelism or angles. This transformation can be represented by a 3x3 matrix H. The perspective transformation is described by the following equation:

$$\begin{pmatrix} x' \\ y' \\ w' \end{pmatrix} = H \begin{pmatrix} x \\ y \\ w \end{pmatrix} \tag{1}$$

(x, y) are the coordinates of a point in the source image and (x', y') are the coordinates of the corresponding point in the destination image. w and w' are scaling factors. H is the 3x3 perspective transformation matrix. The matrix H has 8 degrees of freedom, which can be computed with the knowledge of the correspondences between 4 points in the source image and 4 points in the destination image [15]. We used openCV to annotate the corner points, calculate the transformation matrix, apply the transformation matrix H [16]. Figure 2b visualises the transformed trajectory.

5 Methods of muscle fatigue prediction

Fatigue can be modelled through changing signal parameters of the EMG during muscle contractions [6]. Different signal parameters can be used like the mean absolute value, the AR value, or the zero-crossing rate. In this work, we leverage the median frequency of the EMG signal, a widely used measure for tracking muscle fatigue [7]. To model muscle fatigue over time, the short-term Fourier transform is applied on the windowed EMG. Then, a decrease of median frequency (MF) indicates increasing muscle fatigue [7]. Accordingly, we used the MF to derive a feature for muscle fatigue.

This work aimed to establish a prediction model for the muscle fatigue of a person during climbing. The model takes into account their movement on the wall and trajectory. We compare autoregressive statistical models with ensemble methods like Random Forest (RF) and GB, as well as neural networks, namely MLP and a Long short-term memory (LSTM). We predict muscle fatigue $Y_t = \{y_t, y_{t-1}, \ldots, y_{t-T}\}$ in a recursive manner, where previous observations, endogenous features, predict future observations. Further, we integrate exogenous time-dependent features. The exogenous features include the spatial position of trajectories (x, y), the right arm acceleration in $x-$ and $y-$direction.

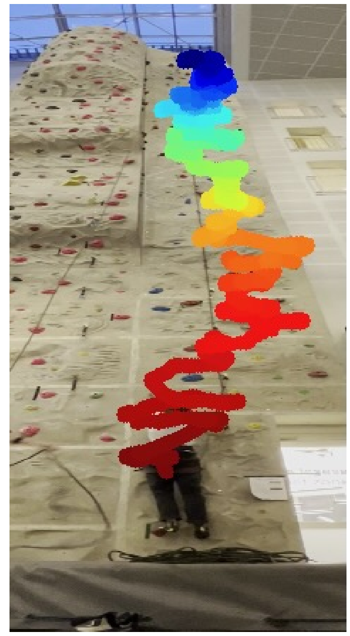
5.1 Autoregressive model

An AR model predicts a current value through a series of previous observations $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}$ [23]. The AR model also allows the integration of exogenous variables by adding them to the regression equation as in Equation 2.

$$y_t = \sum_{i=1}^{p} \phi_i \cdot y_{t-i} + \sum_{i=1}^{n} \sum_{j=1}^{k} \beta_{ij} \cdot x_{jt-i} + \epsilon_t \tag{2}$$

Where, ϕ_i are the coefficients for the endogenous variable Y lagged by i periods. β_{ij} are the coefficients for the j-th exogenous predictor variable lagged by i time periods. x_{jt-i} represents the j-th exogenous predictor variable lagged by i periods. ϵ_t is the error term at time t. Finally, p is the number of lags considered for each variable and k is the number of exogenous predictor variables.

We investigated the autocorrelation and partial autocorrelation plots which allowed us to determine the lag structure of the fatigue signals. The lag structure provides insights into temporal dependencies and determines order of the AR and moving average process.



(a) The annotated trajectory on the original video image. The colours indicate the frame number and show the development of the trajectory over time.

(b) The homography transformed trajectory of the climber onto a 2 dimensional surface. The colours indicate the frame number and show the development of the trajectory over time.

Figure 2: Combined visualisation of the climber's trajectory in both original and transformed views.

According to the the autocorrelation and partial autocorrelation plots, all the feature signals follow a first-order AR process.

AR models assume stationarity, a fundamental property in time series analysis. We call a series weakly stationary if its mean and variance are constant over time and if the covariance between two observations depends only on the time interval between them. Time series without a unit root exhibit such behaviour [23]. The MF of the EMG signal, which indicates muscle fatigue by a linear decrease, inherently exhibits a trend. The presence of a deterministic trend indicates trend stationarity. However, this differs from a series with a unit root, which would mean that the process is inherently non-stationary in its stochastic trend. To effectively address the linear trend, we add a trend term to the AR 2.

5.1.1 AR with expected muscle fatigue. We introduce the expected muscle fatigue given the climbed distance in y-direction, $E(y_t|X_{t,traj_y} \in D_l)$. $X_{t,traj_y}$ is the climbed distance in y direction at time t. We divide the climbed distance into l categories (meters) D_l. The expected value of muscle fatigue, given the distance category, is expressed as,

$$E(y_t \mid X_{t,\text{traj}_y} \in D_l) = \sum_{l=1}^{L} \delta_l D_{lt} \tag{3}$$

For each category, we define D_{lt} as an indicator variable that specifies whether the climbed distance at time t falls within the l-th category. The model includes coefficients δ_l for each indicator variable D_{lt}, which measure the impact of being in the l-th distance category on muscle fatigue. Including the indicator variable into the AR model enables the incorporation of muscle fatigue data from previous climbers, enhancing the prediction of how new climbers might respond under similar conditions. Furthermore, the categories capture unique variance associated with that specific interval of climbed distance, allowing the model to adjust more dynamically to changes in fatigue levels based on the distance climbed.

5.2 Ensemble methods

Ensemble methods improve prediction accuracy and robustness by combining the results of multiple models. For the prediction of muscle fatigue, we use two advanced ensemble methods, namely GB and RF. Both models are capable of handling complex, non-linear data interactions.

GB builds an additive model step by step. It starts with a base learner, in our case a decision tree, and gradually corrects errors by adding new models. This method optimises a differentiable loss function, like the mean square error, which is minimised by the sequential introduction of trees [1, 20].

In RF, multiple decision trees are aggregated to enhance prediction accuracy, with each tree constructed from a randomly sampled subset of the data and features. However, given the sequential nature of time series data, conventional bootstrapping would disrupt the temporal order, leading to inaccurate models. To address this, we employ block bootstrapping, which samples contiguous blocks of $x_t, \ldots, x_{t+o}$, thereby preserving the time structure within each block [5]. The predictive output of the model is the average of the predictions from all trees [1].

Both ensemble methods are trained on with a learning rate of 0.1 and a maximum depth of 3 for each tree. We trained the ensemble methods on block sizes of 10 time steps.

5.3 Deep-learning approach

In this section, we describe the neural network architectures, specifically MLP and LSTM, used for modeling muscle fatigue. The MLP serves as a foundational architecture, effective for smaller datasets and straightforward to train. However, it struggles with sequential data, where LSTM excels. The LSTM architecture is tailored for sequence processing, particularly adept at capturing long-term dependencies in time series data, making it a natural fit for modeling activities such as sleep patterns over extended periods [13, 20].

Both architectures contain dropout layers after each fully linked layer to prevent overfitting. In our implementation, the neural networks take an input time vector for the last 20 time points, spanning a 10-minute interval. The networks were implemented using Pytorch with the Adam optimiser, setting the learning rate at 0.001. We applied early stopping as another technique to prevent overfitting.

5.4 Evaluation metrics

We evaluate the prediction performance using three metrics, the Mean square error (MSE), the RMSE, and the Mean absolute error (MAE). The MSE measures the average of the squared errors. It gives greater weight to large errors due to the squaring element. Moreover, the MSE is used as a loss function and thus offers insights on the minimisation of the loss. RMSE is the square root of the mean of the squared errors. It measures the standard deviation of the prediction errors and, unlike MSE, is on the same scale as the predicted values. Finally, there is MAE, which measures the average magnitude of the errors in a set of predictions without considering their direction. The advantage and the reason why we include this metric is that it is less sensitive to outliers than MSE and is often used when one wants to avoid giving extra weight to larger errors [14].

6 Results

This section contains the results of our prediction of muscle fatigue during climbing. It describes the signal pre-processing, data analysis, visual representations of the data and our results, and a thorough evaluation of the prediction models. We compare the prediction performance of the models presented in Section 5 for short and long time periods based on leave-one-out cross-validation.

6.1 Signal pre-processing

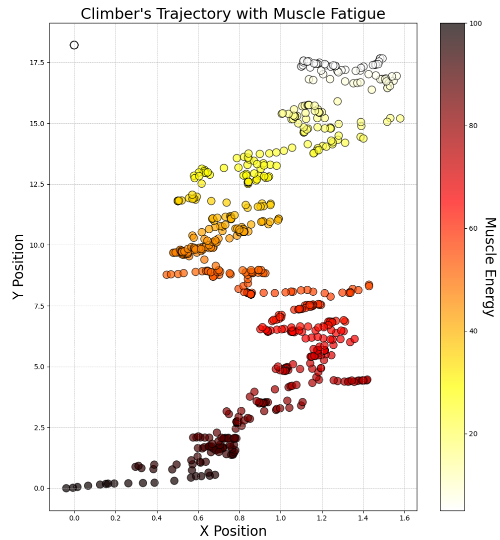
During the experiment we recorded data from EMG, IMU, and video recordings were converted to a 2D motion trajectory. The EMG signal was sampled at $2000Hz$. A notch filter at $50Hz$ suppressed noise from electrical activity (European) and harmonic interference which often contaminate sEMG [18]. The notch filtering was followed by a bandpass filter with the bandwidth of $(20Hz - 450Hz)$ and an order of 4 [21]. The IMU was sampled at $50Hz$. We filtered the IMU with a bandpass filter with the bandwidth of $(0.1Hz - 50Hz)$. The filter order was set to 4. The video is recorded at $30fps$ accordingly, the trajectory is sampled at $30Hz$. As described in Section 4, the annotation was live and not performed frame by frame. Missing frame annotations were forward-filled with the last position.

6.2 Data analysis and visualisation

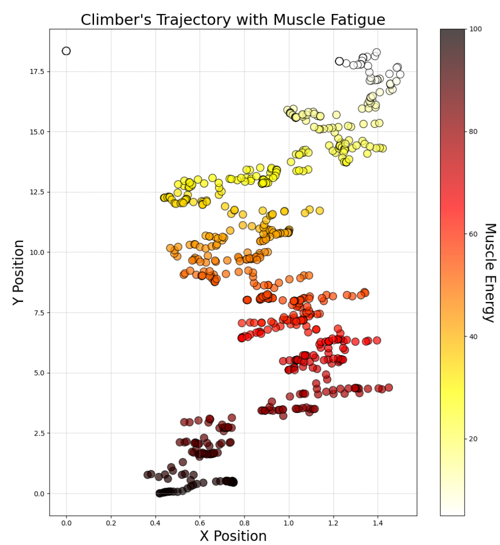
We conducted a study with 20 recreational climbers who climbed the same route under two different conditions: top rope and lead climbing. This resulted in a total of 40 recorded climbs. The cohort consisted of 12 men and 8 women with an average age of 31 (±7.8) between 32 and 50 years. Most of the participants had up to four years of climbing experience and dedicated an average of one hour per week to climbing and one hour to bouldering (climbing without the use of a rope for protection). In addition to the climbing trajectory, the EMG of the FDS and right arm movement of each participant was recorded. In Figure 3, we visualise the linear increase of muscle fatigue measured by the linear decrease of the median frequency of the EMG signal. With the derived information, we can compare individual performances as well as differences between climbing styles.

This study is primarily concerned with developing a prediction model that can predict a climber's muscle fatigue. As described in Section 5, our approach extends the traditional AR model by integrating the estimated expected muscle fatigue based on the vertical distance climbed. Figure 4 illustrates the density of the median frequency over the altitude meters. The densities visually represent the distribution of the median frequency of the EMG signal. The figure underlines the linear decrease over all participants. The shape of the density changes only slightly over time and remains relatively constant. By linking the observed EMG median frequencies to the predicted fatigue levels, the model uses the expected EMG median frequency of previous climbers to improve the accuracy and reliability of its predictions.

In Section 5, we introduced a stationarity term and the difference between unit-root and trend stationarity. The time series need to be tested for unit-root stationarity. MF signal of each participant was tested for stationarity with the Augmented-Dickey-Fuller test [4]. The null hypothesis of the augmented Dickey-Fuller test is non-stationarity. Only 5 of the 40 time series were found to be non-stationary. Given the limited size of our sample, excluding these five series would substantially reduce the available data, potentially affecting the statistical power and generalisability of our results. The results of the augmented Dickey-Fuller test are provided in the supplementary material.



(a) Top rope climbing



(b) Lead climbing

Figure 3: The trajectories show the increase in muscle fatigue over the course of the ascent. The lower the muscle energy, the higher the muscle fatigue. The two diagrams compare the lead and top rope ascent of participant 6 as an example. To illustrate this, we modelled muscle fatigue increase in the FDS linearly.

6.3 Muscle fatigue prediction

We predicted muscle fatigue in the form of the MF of the EMG signal recorded by the FDS of a climber. We compared four prediction models, namely a statistical AR model, our proposed AR model with expected muscle fatigue given climbed distance in y-direction, a MLP, and a LSTM.

Due to the small sample size of climbs, we trained the models based on a leave-one-out cross-validation. This means that we trained the models on 39 climbs and predicted muscle fatigue for one. This was repeated 40 times and the resulting error metrics were averaged. Hence, the models predicted the full course of muscle fatigue of a climber given the other climbers. We did not differentiate between lead and top rope approaches during model training. As described in Section 3, the climbers mimicked the rope clipping movements of the lead climber during the top rope approach to make both approaches comparable. Accordingly, we would expect a comparable fatigue development. Whether there are possible differences due to mental stress or technique is not the subject of this paper.

Additionally, we compared the predictive performance of the models for a short and long period. Regarding the short period, the

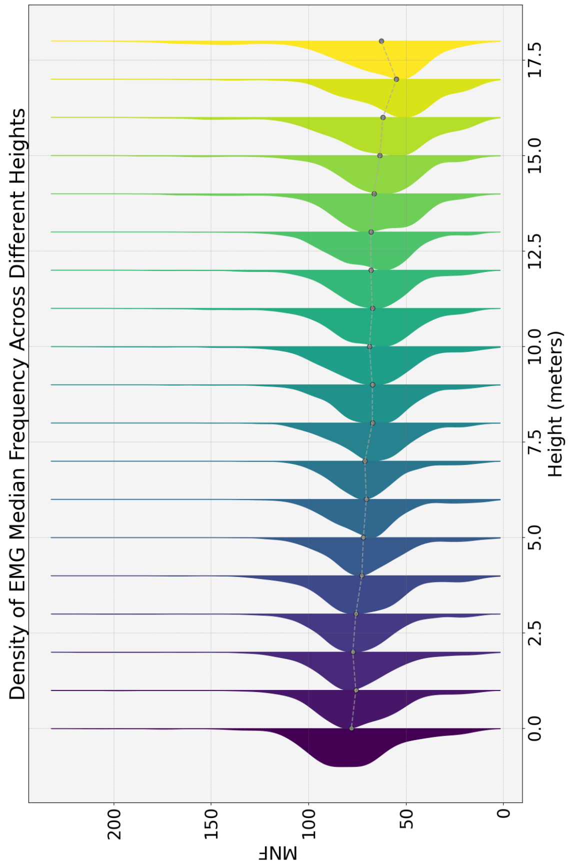


Figure 4: The density of the EMG's median frequency across meters of height. Each density plot is connected through its median. The connecting line shows a decrease in median frequency with increasing height.

models predict the next time step y_{t+1}. As the time series is sampled at 4Hz the short period predicts the next 250 ms. The long period predicts the 20th time step y_{t+20} which corresponds to 5 seconds.

Besides, using the MF of the EMG as the endogenous feature, we also included exogenous variables. As described in 3, the sensor used also includes an IMU, from which we could derive the acceleration of the arm in x- and y-directions. Moreover, we included the climbers' spatial (x, y) position on the wall.

We compared the predictive performance of the autoregressive model, the ensemble methods, and the neural networks. Table 1 provides the mean and standard deviation derived from the 5-fold cross-validation of the performance metrics introduced in Section 5. The table compares all six models for the short and long-term forecasts. First of all, it seems that the more complex and non-linear neural networks and ensemble methods achieve a better predictive performance than the linear autoregressive models except the LSTM. Including the expected fatigue into the AR model decreases all three metrics drastically and improves the model compared to the linear model. This accounts for the short- and long-term predictions. Moreover, not only the mean performance metrics are decreased but the standard deviation. Among the non-linear models, the MLP consistently outperformed across all metrics for long-term predictions and achieved the lowest MSE and RMSE for short-term predictions. The GB recorded the lowest MAE for short-term forecasts. The MLP and GB not only perform better on average but also show relatively lower variability in their performance through a low standard deviation.

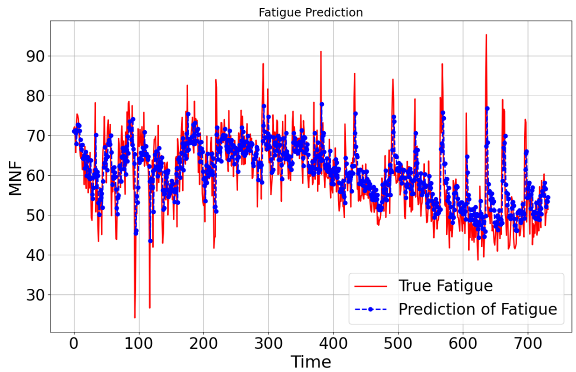
Figure 5 shows and compares the short- and long-term prediction of muscle fatigue predicted by a MLP. The MLP results are presented as it is the best-performing model. The short-term prediction follows the progression of muscle fatigue very closely as we predicted a quarter of a second into the future. It shows the potential for capturing the immediate fluctuations of muscle fatigue. The five-second long-term prediction manages to model the negative trend of fatigue and muscle activation, however, the predictions were delayed in time. While it does not affect the modelling of the general progression of muscle fatigue, it could be crucial in modelling the progression of fatigue in greater detail.

Table 1: The comparative results of predicting muscle fatigue for short and long term forecasts, with SD as a percentage of the mean.

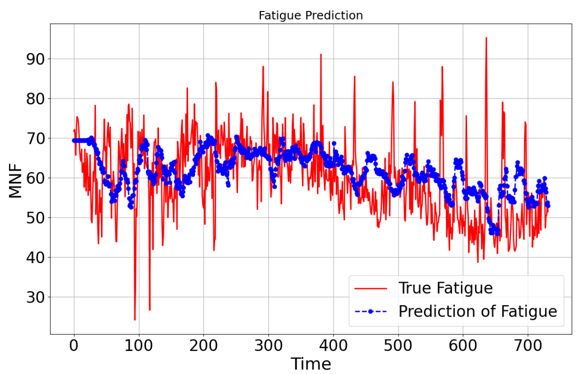
Model	MSE (±SD%)	RMSE (±SD%)	MAE (±SD%)
	Short Term Forecasts		
AR	284.13 ± *61.41%*	16.24 ± *28.11%*	11.14 ± *26.34%*
AR (Exp. Fat.)	278.02 ± *61.71%*	16.05 ± *28.42%*	11.09 ± *26.89%*
ANN	**245.14 ± *58.61%***	**15.09 ± *27.94%***	10.45 ± *29.57%*
LSTM	337.35 ± *75.86%*	17.44 ± *33.53%*	12.71 ± *35.81%*
GBM	246.42 ± *61.32%*	15.09 ± *29.02%*	**10.42 ± *31.21%***
RF	273.43 ± *62.23%*	15.90 ± *29.06%*	11.19 ± *30.12%*
	Long Term Forecasts		
AR	430.32 ± *67.64%*	19.75 ± *32.30%*	14.21 ± *35.11%*
AR (Exp. Fat.)	417.51 ± *68.37%*	19.44 ± *32.60%*	14.04 ± *35.19%*
ANN	**342.35 ± *66.72%***	**17.63 ± *32.14%***	**12.57 ± *35.49%***
LSTM	428.46 ± *76.13%*	19.57 ± *34.70%*	14.54 ± *38.30%*
GBM	345.46 ± *70.92%*	17.65 ± *33.25%*	12.57 ± *37.09%*
RF	366.54 ± *70.92%*	18.20 ± *32.81%*	13.13 ± *36.02%*

7 Discussion

In this paper, we presented the predictive modelling of muscle fatigue of climbers during climbing lead and top rope. To do so, we conducted a multi-modal experiment, which combined sensor data and a video trajectory of 20 climbers climbing on lead and top rope. The sensor data was placed on the FDS to record the EMG signal and to capture muscle fatigue of the climbers. To the best of our knowledge, we are the first to have measured EMG and thus muscle activity during climbing and combined it with climbers' trajectories.



(a) Short term muscle fatigue forecast



(b) Long term muscle fatigue forecast

Figure 5: Comparative results of muscle fatigue predictions using an MLP model. Panel (a) displays the short-term predictions versus actual muscle fatigue measurements, while Panel (b) shows the long-term predictions. The blue lines indicate the predicted fatigue and the red lines represent the actual measured fatigue over time. The muscle fatigue is measured as the MF of the EMG signal, measured during climbing.

As a baseline, we modelled the muscle fatigue as a linear autoregressive model of order one. To improve the statistical model, we included the expected fatigue in the model. The proposed autoregressive model improved forecasting performance over the normal autoregressive model. Moreover, the included expected muscle fatigue reduces the standard deviation of the performance metrics. This means that the included term improves the stability of the predictions. With a reduced standard deviation of the leave-one-out cross-validation, the model's predictions are consistently closer to the actual values across different subsets of the data.

As discussed in Section 6.3, the experimental results clearly show that non-linear and more complex neural networks and ensemble methods outperform linear models in predicting muscle fatigue. This observation is particularly relevant when considering that a negative linear trend is associated with muscle fatigue. However, the actual time series data for muscle fatigue show repeated patterns of activation of the muscles, which necessitates the use of more sophisticated models.

The complexity of muscle activation patterns, which vary not only due to physiological differences between climbers but also due to different climbing styles and methods, poses major challenges for simpler predictive models. While autoregressive models are inherently capable of capturing periodic patterns, the irregular nature of these activations in climbing scenarios limits their effectiveness. The temporal dynamics of climbing are not strictly periodic but are influenced by intermittent and climber-specific factors that are better captured by models that can learn non-linear dependencies and complex interactions within the data.

Surprisingly, despite its ability to effectively model sequence data, the LSTM does not perform as expected in this study. A plausible explanation for this could be the inherent model complexity of LSTM, which may not be well suited for the relatively small dataset available in this case study. This discrepancy highlights a critical aspect of model selection in machine learning for physiological data: the balance between model complexity and the amount and variety of data available.

8 Conclusions

Predicting future muscle fatigue not only enhances our understanding of physical endurance but also plays a crucial role in preventing potential falls and optimising rest times for climbers. While the MLP and GB models performed better than conventional linear models in our studies, it is important to consider the practical applications of these results.

Linear models, despite their lower prediction accuracy, offer advantages in terms of computational simplicity and real-time application. These models are lightweight enough that they can be implemented on portable devices and provide climbers with real-time predictions of muscle fatigue. This capability can be incredibly valuable in real-world climbing scenarios, as immediate feedback on fatigue levels could help climbers make informed decisions about when to rest or whether to continue climbing.

Even though the results of this study are promising, some limitations must be taken into account. First, the relatively small dataset of 40 climbing sessions on one route limits the generalisability of our results. In addition, the analysis was limited to the physiological signals of the right arm, which may not fully reflect general muscle fatigue. The climbing trajectories were manually labeled, which could be automated through advanced technologies such as OpenPose to improve the efficiency of data collection.

To extend the foundations of this study, future research should aim to expand the dataset to include more different climbing routes and both arms to better generalise the prediction of fatigue under different climbing conditions. The inclusion of additional physiological markers, such as heart rate and heart rate variability, could deepen our understanding of systemic responses to physical exertion and improve the predictive accuracy of our models. A critical examination of the signals that most effectively predict muscle fatigue is also essential. Determining the predictive power of individual features will optimise the efficiency of the model, and exploring other potential biomarkers, such as metabolic markers accessible via non-invasive sensors, could pave the way for the development of user-friendly fatigue assessment tools. The ultimate goal of this line of research is to apply these refined models to the development of algorithms suitable for integration into fitness wearables.

9 Code availability

The code used in this study is contained in the publicly accessible GitHub repository [2].

10 Data availability

The dataset supporting the findings of this study is available in a repository on the Open Science Framework [3].

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